

Linux Training Guide

Willrock Herrera

UMGC – CMIT 291

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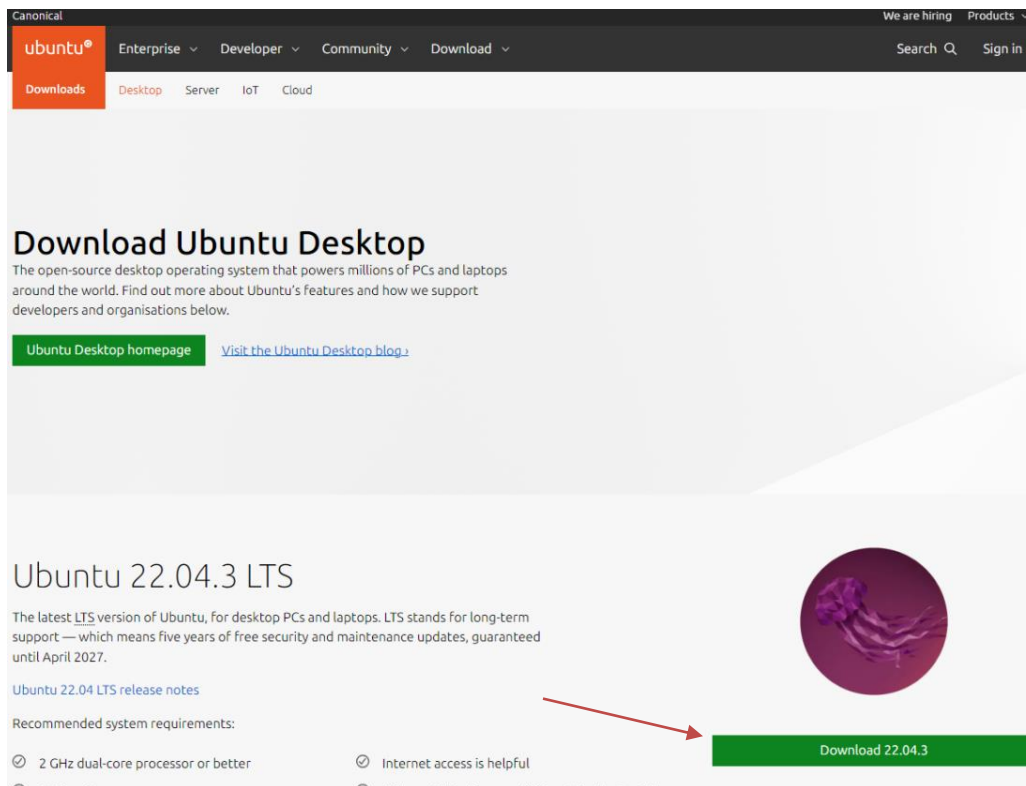
Linux Training Guide

Introduction

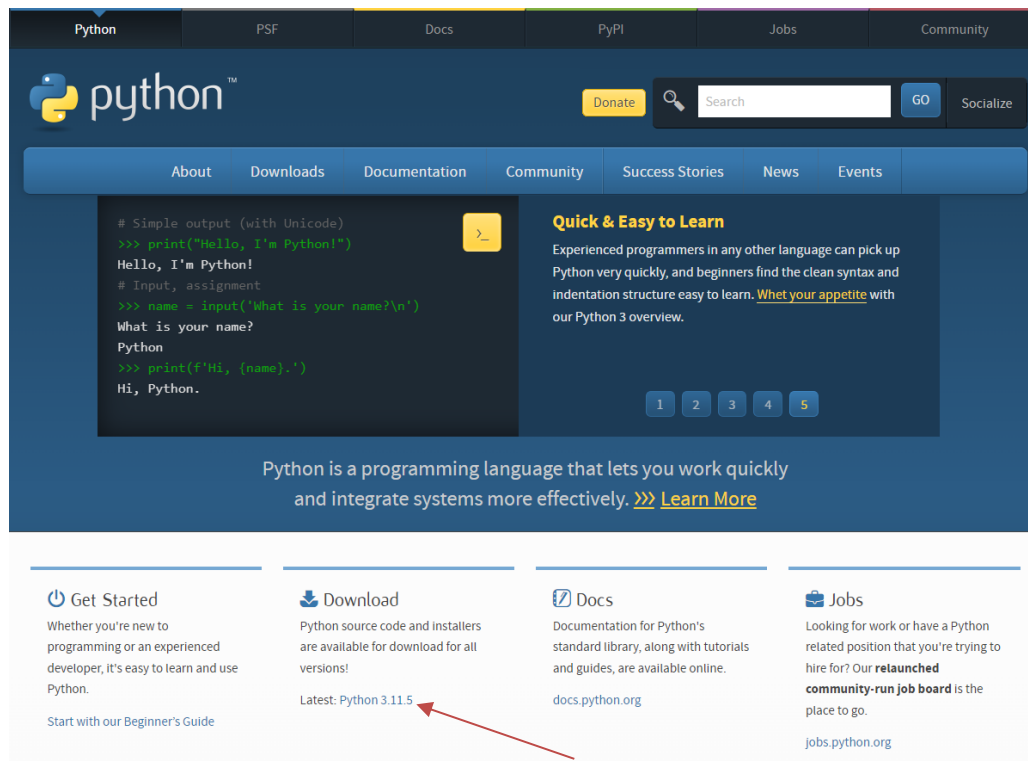
Thank you so much for deciding to go through this tremendous move. Those of us at Go2Linux are super excited to join Faster Computing on this journey. Should the need arise, we will always be available as a resource and for future needs. Now let us begin our training guide.

Part 1

Step 1: We will begin by downloading the latest version of Ubuntu at <https://ubuntu.com/download/desktop>.



Step 2: We will need to download the latest version of python from <https://www.python.org/> and install it as soon as possible by running the application and following the prompts.



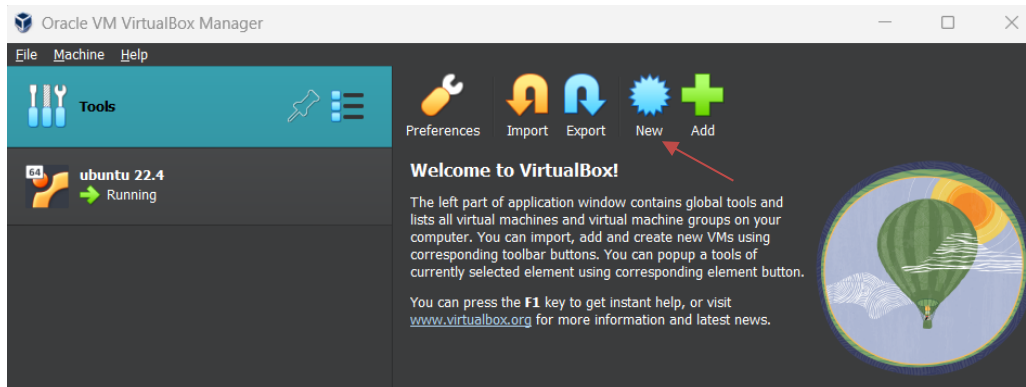
Step 3: After the .iso file for Ubuntu has finished downloading, we will continue with downloading the Virtual Machine (VM) “VirtualBox” by Oracle at

<https://www.virtualbox.org/wiki/Downloads>



Step 4: Run the VirtualBox app to begin the installation wizard and follow the prompts until it opens. After successfully installing VirtualBox, open the application so we can begin installing the Ubuntu Operating System.

Step 5: When we open the Oracle VirtualBox app we will want to select “New” to create a new VM.



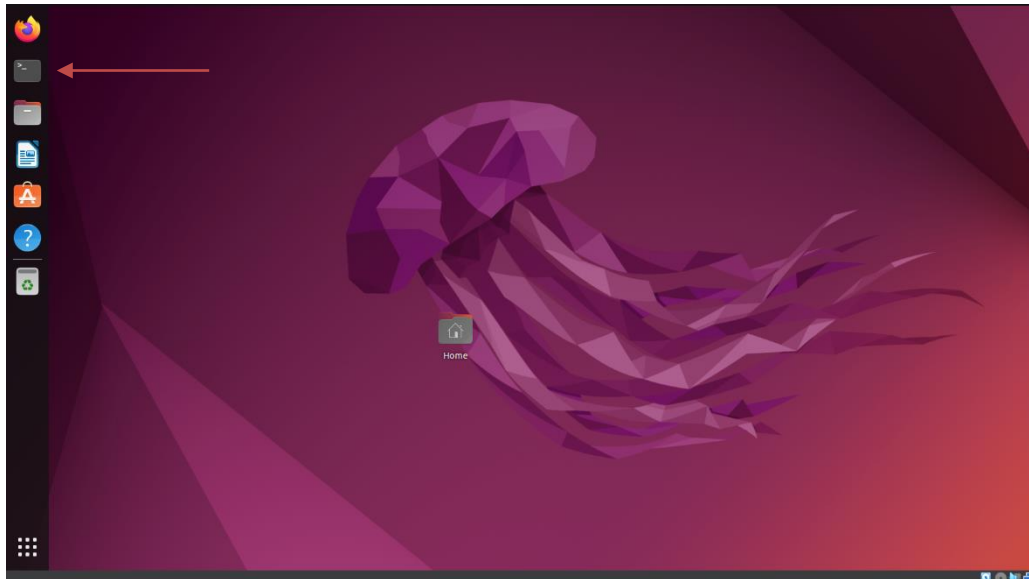
Step 6: We will then get a “Create Virtual Machine” window that prompts us with steps we must complete.

- Name: create a name for your virtual machine
- Folder: leave with the default destination
- ISO image: you must select the Ubuntu file previously downloaded for the VM to boot up the first time with Ubuntu.
- Edition: remains blank
- Type: Should auto-populate after typing in ubuntu for “Name”
- Version: Verify that “Ubuntu (64-bit)” is selected. And then clicking next.
- On the “Unattended Guest OS Install Setup” window, verify username and password are prefilled to “vboxuser” and “changeme” respectfully. Also, verify Hostname does not have any spaces. Lastly ensure the “Guest Additions” checkbox is selected and click next.
- On the “Hardware” window, ensure Base Memory is at a minimum of 2048 MB, preferably 5 GB, and Processors at a minimum of 2 CPUs, preferably 8 CPUs, and click next.
- On the “Virtual Hard disk” window, select the “Create a Virtual Hard Disk Now” radio button and set to a minimum of 25 GB, preferably 100GB, and click next.
- On the “Summary” window click next and on the last window click finish.
- The OS will automatically open and will be installing the OS. Once it is finished, you’ll be ready to start using the system.

Part 2

Demonstrate command-line operations that will provide the following information:

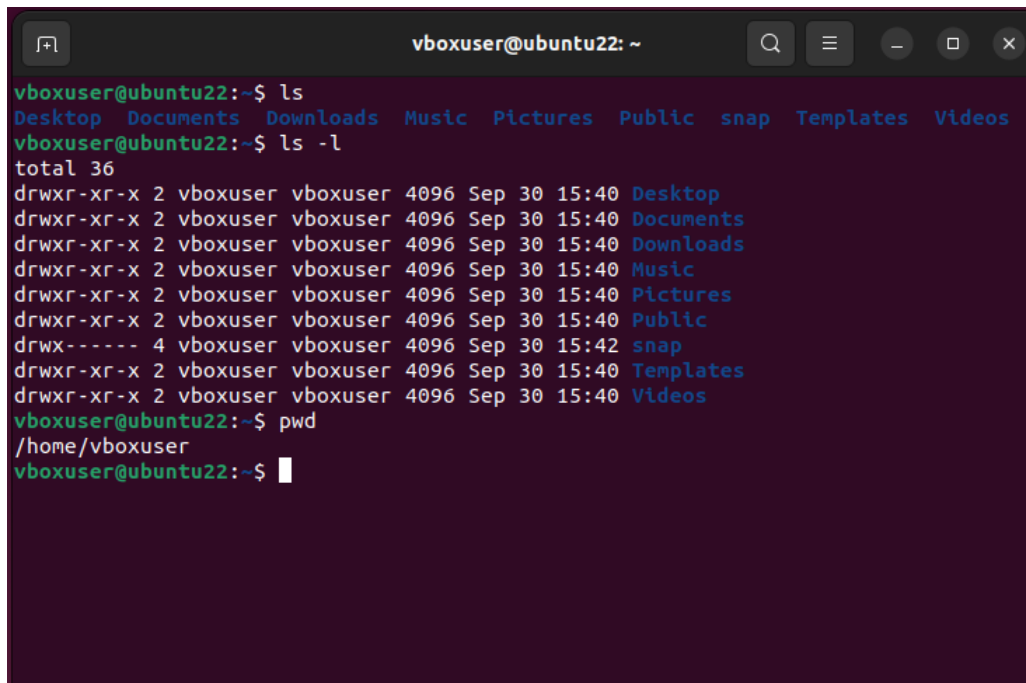
- First, you must open the “Terminal” application.



- A listing of files in a directory and common file attribute.
 - The command “ls” will display a full list of all files in the system and “ls -l” will display the same with more detailed information.

```
vboxuser@ubuntu22: ~  
vboxuser@ubuntu22:~$ ls  
Desktop Documents Downloads Music Pictures Public snap Templates Videos  
vboxuser@ubuntu22:~$ ls -l  
total 36  
drwxr-xr-x 2 vboxuser vboxuser 4096 Sep 30 15:40 Desktop  
drwxr-xr-x 2 vboxuser vboxuser 4096 Sep 30 15:40 Documents  
drwxr-xr-x 2 vboxuser vboxuser 4096 Sep 30 15:40 Downloads  
drwxr-xr-x 2 vboxuser vboxuser 4096 Sep 30 15:40 Music  
drwxr-xr-x 2 vboxuser vboxuser 4096 Sep 30 15:40 Pictures  
drwxr-xr-x 2 vboxuser vboxuser 4096 Sep 30 15:40 Public  
drwx----- 4 vboxuser vboxuser 4096 Sep 30 15:42 snap  
drwxr-xr-x 2 vboxuser vboxuser 4096 Sep 30 15:40 Templates  
drwxr-xr-x 2 vboxuser vboxuser 4096 Sep 30 15:40 Videos  
vboxuser@ubuntu22:~$
```

- The current directory (hint: also known as the present working directory).
 - The command “pwd” will display the current working directory.



```
vboxuser@ubuntu22: ~  
vboxuser@ubuntu22:~$ ls  
Desktop Documents Downloads Music Pictures Public snap Templates Videos  
vboxuser@ubuntu22:~$ ls -l  
total 36  
drwxr-xr-x 2 vboxuser vboxuser 4096 Sep 30 15:40 Desktop  
drwxr-xr-x 2 vboxuser vboxuser 4096 Sep 30 15:40 Documents  
drwxr-xr-x 2 vboxuser vboxuser 4096 Sep 30 15:40 Downloads  
drwxr-xr-x 2 vboxuser vboxuser 4096 Sep 30 15:40 Music  
drwxr-xr-x 2 vboxuser vboxuser 4096 Sep 30 15:40 Pictures  
drwxr-xr-x 2 vboxuser vboxuser 4096 Sep 30 15:40 Public  
drwx----- 4 vboxuser vboxuser 4096 Sep 30 15:42 snap  
drwxr-xr-x 2 vboxuser vboxuser 4096 Sep 30 15:40 Templates  
drwxr-xr-x 2 vboxuser vboxuser 4096 Sep 30 15:40 Videos  
vboxuser@ubuntu22:~$ pwd  
/home/vboxuser  
vboxuser@ubuntu22:~$
```

- Create a file, then copy it to a different directory.
 - We will start by creating two directories simultaneously from the terminal, titling the first as “direct_one” and making the second “direct_two” as follows (directories can be labeled anything desired): “mkdir direct_one direct_two”

```
vboxuser@ubuntu22: ~  
vboxuser@ubuntu22:~$ ls  
Desktop Documents Downloads Music Pictures Public snap Templates Videos  
vboxuser@ubuntu22:~$ ls -l  
total 36  
drwxr-xr-x 2 vboxuser vboxuser 4096 Sep 30 15:40 Desktop  
drwxr-xr-x 2 vboxuser vboxuser 4096 Sep 30 15:40 Documents  
drwxr-xr-x 2 vboxuser vboxuser 4096 Sep 30 15:40 Downloads  
drwxr-xr-x 2 vboxuser vboxuser 4096 Sep 30 15:40 Music  
drwxr-xr-x 2 vboxuser vboxuser 4096 Sep 30 15:40 Pictures  
drwxr-xr-x 2 vboxuser vboxuser 4096 Sep 30 15:40 Public  
drwx----- 4 vboxuser vboxuser 4096 Sep 30 15:42 snap  
drwxr-xr-x 2 vboxuser vboxuser 4096 Sep 30 15:40 Templates  
drwxr-xr-x 2 vboxuser vboxuser 4096 Sep 30 15:40 Videos  
vboxuser@ubuntu22:~$ pwd  
/home/vboxuser  
vboxuser@ubuntu22:~$ mkdir direct_one direct_two  
vboxuser@ubuntu22:~$ ls  
Desktop direct_two Downloads Pictures snap Videos  
direct_one Documents Music Public Templates  
vboxuser@ubuntu22:~$
```

- Next, we will prepare to save a file inside one of the directories by using the “cd” command which changes the current directory. Input the following the command: “cd direct_one”

```
vboxuser@ubuntu22: ~/direct_one  
vboxuser@ubuntu22:~$ ls  
Desktop Documents Downloads Music Pictures Public snap Templates Videos  
vboxuser@ubuntu22:~$ ls -l  
total 36  
drwxr-xr-x 2 vboxuser vboxuser 4096 Sep 30 15:40 Desktop  
drwxr-xr-x 2 vboxuser vboxuser 4096 Sep 30 15:40 Documents  
drwxr-xr-x 2 vboxuser vboxuser 4096 Sep 30 15:40 Downloads  
drwxr-xr-x 2 vboxuser vboxuser 4096 Sep 30 15:40 Music  
drwxr-xr-x 2 vboxuser vboxuser 4096 Sep 30 15:40 Pictures  
drwxr-xr-x 2 vboxuser vboxuser 4096 Sep 30 15:40 Public  
drwx----- 4 vboxuser vboxuser 4096 Sep 30 15:42 snap  
drwxr-xr-x 2 vboxuser vboxuser 4096 Sep 30 15:40 Templates  
drwxr-xr-x 2 vboxuser vboxuser 4096 Sep 30 15:40 Videos  
vboxuser@ubuntu22:~$ pwd  
/home/vboxuser  
vboxuser@ubuntu22:~$ mkdir direct_one direct_two  
vboxuser@ubuntu22:~$ ls  
Desktop direct_two Downloads Pictures snap Videos  
direct_one Documents Music Public Templates  
vboxuser@ubuntu22:~$ cd direct_one  
vboxuser@ubuntu22:~/direct_one$ pwd  
/home/vboxuser/direct_one  
vboxuser@ubuntu22:~/direct_one$
```

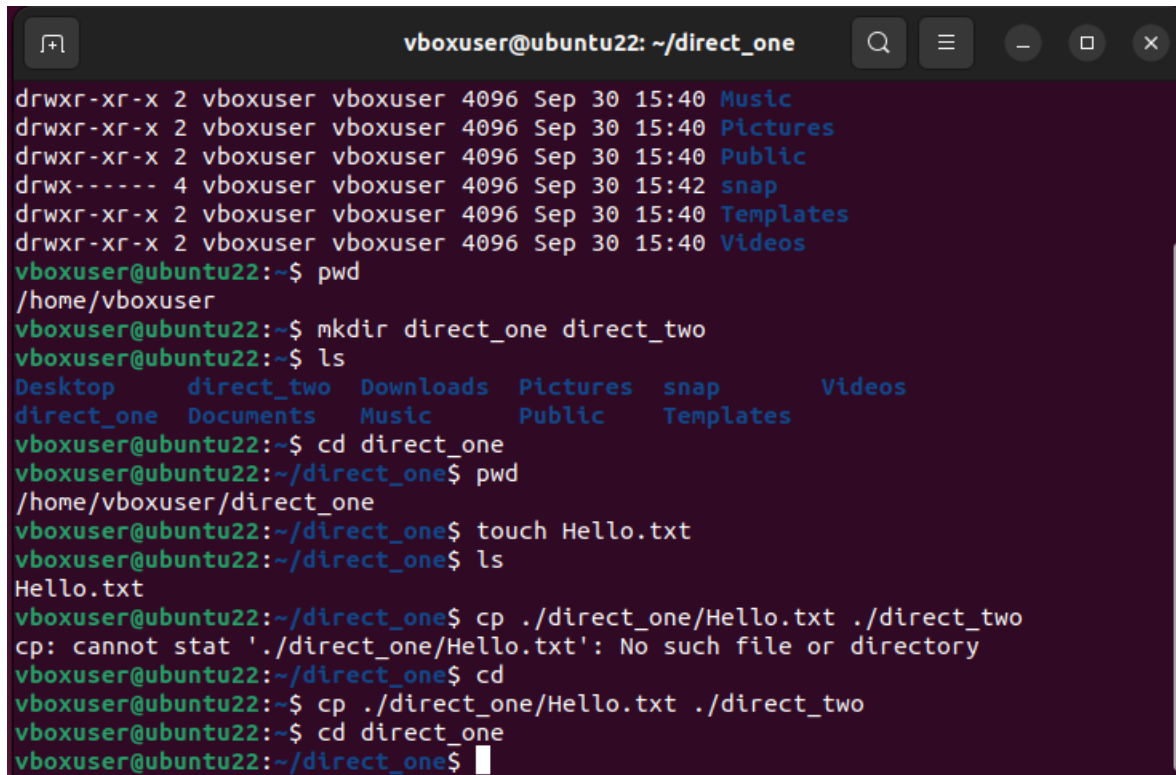
- To create a brand-new empty file, we will use the “touch” command inside the terminal and naming the file “Hello” as follows: “touch Hello.txt”

```
vboxuser@ubuntu22: ~/direct_one
vboxuser@ubuntu22:~$ ls -l
total 36
drwxr-xr-x 2 vboxuser vboxuser 4096 Sep 30 15:40 Desktop
drwxr-xr-x 2 vboxuser vboxuser 4096 Sep 30 15:40 Documents
drwxr-xr-x 2 vboxuser vboxuser 4096 Sep 30 15:40 Downloads
drwxr-xr-x 2 vboxuser vboxuser 4096 Sep 30 15:40 Music
drwxr-xr-x 2 vboxuser vboxuser 4096 Sep 30 15:40 Pictures
drwxr-xr-x 2 vboxuser vboxuser 4096 Sep 30 15:40 Public
drwx----- 4 vboxuser vboxuser 4096 Sep 30 15:42 snap
drwxr-xr-x 2 vboxuser vboxuser 4096 Sep 30 15:40 Templates
drwxr-xr-x 2 vboxuser vboxuser 4096 Sep 30 15:40 Videos
vboxuser@ubuntu22:~$ pwd
/home/vboxuser
vboxuser@ubuntu22:~$ mkdir direct_one direct_two
vboxuser@ubuntu22:~$ ls
Desktop  direct_two  Downloads  Pictures  snap      Videos
direct_one Documents  Music      Public    Templates
vboxuser@ubuntu22:~$ cd direct_one
vboxuser@ubuntu22:~/direct_one$ pwd
/home/vboxuser/direct_one
vboxuser@ubuntu22:~/direct_one$ touch Hello.txt
vboxuser@ubuntu22:~/direct_one$ ls
Hello.txt
vboxuser@ubuntu22:~/direct_one$
```

- The “touch” command also allows for creating multiple files at once by adding a space between each file name as such: touch file_one.txt file_two.txt file_three.txt file_four.txt” etc
- Lastly we will copy this file into our second directory with the “cp” command from the original directory: “cd” to return to original directory and then “cp ./direct_one/Hello.txt ./direct_two”. The user can copy multiple files at once similarly to how we can create multiple files by inputting the source files first and adding the destination directory at the end.

```
vboxuser@ubuntu22: ~
drwxr-xr-x 2 vboxuser vboxuser 4096 Sep 30 15:40 Downloads
drwxr-xr-x 2 vboxuser vboxuser 4096 Sep 30 15:40 Music
drwxr-xr-x 2 vboxuser vboxuser 4096 Sep 30 15:40 Pictures
drwxr-xr-x 2 vboxuser vboxuser 4096 Sep 30 15:40 Public
drwx----- 4 vboxuser vboxuser 4096 Sep 30 15:42 snap
drwxr-xr-x 2 vboxuser vboxuser 4096 Sep 30 15:40 Templates
drwxr-xr-x 2 vboxuser vboxuser 4096 Sep 30 15:40 Videos
vboxuser@ubuntu22:~$ pwd
/home/vboxuser
vboxuser@ubuntu22:~$ mkdir direct_one direct_two
vboxuser@ubuntu22:~$ ls
Desktop  direct_two  Downloads  Pictures  snap      Videos
direct_one Documents  Music      Public    Templates
vboxuser@ubuntu22:~$ cd direct_one
vboxuser@ubuntu22:~/direct_one$ pwd
/home/vboxuser/direct_one
vboxuser@ubuntu22:~/direct_one$ touch Hello.txt
vboxuser@ubuntu22:~/direct_one$ ls
Hello.txt
vboxuser@ubuntu22:~/direct_one$ cp ./direct_one/Hello.txt ./direct_two
cp: cannot stat './direct_one/Hello.txt': No such file or directory
vboxuser@ubuntu22:~/direct_one$ cd
vboxuser@ubuntu22:~$ cp ./direct_one/Hello.txt ./direct_two
vboxuser@ubuntu22:~$
```

- Create a second file and move it to a different directory.
 - First, please ensure that we are in direct_one. If not, we can get there by changing directory as follows: “cd” to get to original directory and then inputting “cd direct_one” to be in directory one.



```
vboxuser@ubuntu22: ~/direct_one
drwxr-xr-x 2 vboxuser vboxuser 4096 Sep 30 15:40 Music
drwxr-xr-x 2 vboxuser vboxuser 4096 Sep 30 15:40 Pictures
drwxr-xr-x 2 vboxuser vboxuser 4096 Sep 30 15:40 Public
drwx----- 4 vboxuser vboxuser 4096 Sep 30 15:42 snap
drwxr-xr-x 2 vboxuser vboxuser 4096 Sep 30 15:40 Templates
drwxr-xr-x 2 vboxuser vboxuser 4096 Sep 30 15:40 Videos
vboxuser@ubuntu22:~$ pwd
/home/vboxuser
vboxuser@ubuntu22:~$ mkdir direct_one direct_two
vboxuser@ubuntu22:~$ ls
Desktop    direct_two  Downloads  Pictures    snap        Videos
direct_one Documents   Music      Public      Templates
vboxuser@ubuntu22:~$ cd direct_one
vboxuser@ubuntu22:~/direct_one$ pwd
/home/vboxuser/direct_one
vboxuser@ubuntu22:~/direct_one$ touch Hello.txt
vboxuser@ubuntu22:~/direct_one$ ls
Hello.txt
vboxuser@ubuntu22:~/direct_one$ cp ./direct_one/Hello.txt ./direct_two
cp: cannot stat './direct_one/Hello.txt': No such file or directory
vboxuser@ubuntu22:~/direct_one$ cd
vboxuser@ubuntu22:~$ cp ./direct_one/Hello.txt ./direct_two
vboxuser@ubuntu22:~$ cd direct_one
vboxuser@ubuntu22:~/direct_one$
```

- To create a second file, we will follow the same process as before. In the terminal input the following: “touch World.txt”

```
vboxuser@ubuntu22: ~/direct_one
drwx----- 4 vboxuser vboxuser 4096 Sep 30 15:42 snap
drwxr-xr-x 2 vboxuser vboxuser 4096 Sep 30 15:40 Templates
drwxr-xr-x 2 vboxuser vboxuser 4096 Sep 30 15:40 Videos
vboxuser@ubuntu22:~$ pwd
/home/vboxuser
vboxuser@ubuntu22:~$ mkdir direct_one direct_two
vboxuser@ubuntu22:~$ ls
Desktop  direct_two  Downloads  Pictures  snap      Videos
direct_one Documents  Music      Public    Templates
vboxuser@ubuntu22:~$ cd direct_one
vboxuser@ubuntu22:~/direct_one$ pwd
/home/vboxuser/direct_one
vboxuser@ubuntu22:~/direct_one$ touch Hello.txt
vboxuser@ubuntu22:~/direct_one$ ls
Hello.txt
vboxuser@ubuntu22:~/direct_one$ cp ./direct_one/Hello.txt ./direct_two
cp: cannot stat './direct_one/Hello.txt': No such file or directory
vboxuser@ubuntu22:~/direct_one$ cd
vboxuser@ubuntu22:~$ cp ./direct_one/Hello.txt ./direct_two
vboxuser@ubuntu22:~$ cd direct_one
vboxuser@ubuntu22:~/direct_one$ touch World.txt
vboxuser@ubuntu22:~/direct_one$ ls
Hello.txt  World.txt
vboxuser@ubuntu22:~/direct_one$
```

- Lastly, we will move the file World.txt from direct_one to direct_two by using the “mv” command as follows: “mv World.txt /home/vboxuser/direct_two. If we are in the original directory we would use “mv ./direct_one/World.txt ./direct_two

```
vboxuser@ubuntu22: ~/direct_two
vboxuser@ubuntu22:~/direct_one$ mv World.txt ./direct_two
vboxuser@ubuntu22:~/direct_one$ cd
vboxuser@ubuntu22:~$ ls
Desktop  direct_two  Downloads  Pictures  snap      Videos
direct_one Documents  Music      Public    Templates
vboxuser@ubuntu22:~$ cd direct_two
vboxuser@ubuntu22:~/direct_two$ ls
Hello.txt
vboxuser@ubuntu22:~/direct_two$ cd direct_one
bash: cd: direct_one: No such file or directory
vboxuser@ubuntu22:~/direct_two$ cd
vboxuser@ubuntu22:~$ cd direct_one
vboxuser@ubuntu22:~/direct_one$ ls
Hello.txt  World.txt
vboxuser@ubuntu22:~/direct_one$ cd
vboxuser@ubuntu22:~$ cd direct_one
vboxuser@ubuntu22:~/direct_one$ ls
Hello.txt  World.txt
vboxuser@ubuntu22:~/direct_one$ mv World.txt /home/vboxuser/direct_two
vboxuser@ubuntu22:~/direct_one$ cd
vboxuser@ubuntu22:~$ cd direct_two
vboxuser@ubuntu22:~/direct_two$ ls
Hello.txt  World.txt
vboxuser@ubuntu22:~/direct_two$
```

```

vboxuser@ubuntu22:~$ cd direct_two
vboxuser@ubuntu22:~/direct_two$ ls
Hello.txt
vboxuser@ubuntu22:~/direct_two$ pwd
/home/vboxuser/direct_two
vboxuser@ubuntu22:~/direct_two$ cd
vboxuser@ubuntu22:~$ mv ./direct_two/World.txt ./direct_one
vboxuser@ubuntu22:~$ cd direct_one
vboxuser@ubuntu22:~/direct_one$ ls
Hello.txt World.txt
vboxuser@ubuntu22:~/direct_one$ cd
vboxuser@ubuntu22:~$ cd direct_two
vboxuser@ubuntu22:~/direct_two$ ls
Hello.txt
vboxuser@ubuntu22:~/direct_two$ cd
vboxuser@ubuntu22:~$ mv ./direct_one/World.txt ./direct_two
vboxuser@ubuntu22:~$ cd direct_two
vboxuser@ubuntu22:~/direct_two$ ls
Hello.txt World.txt
vboxuser@ubuntu22:~/direct_two$

```

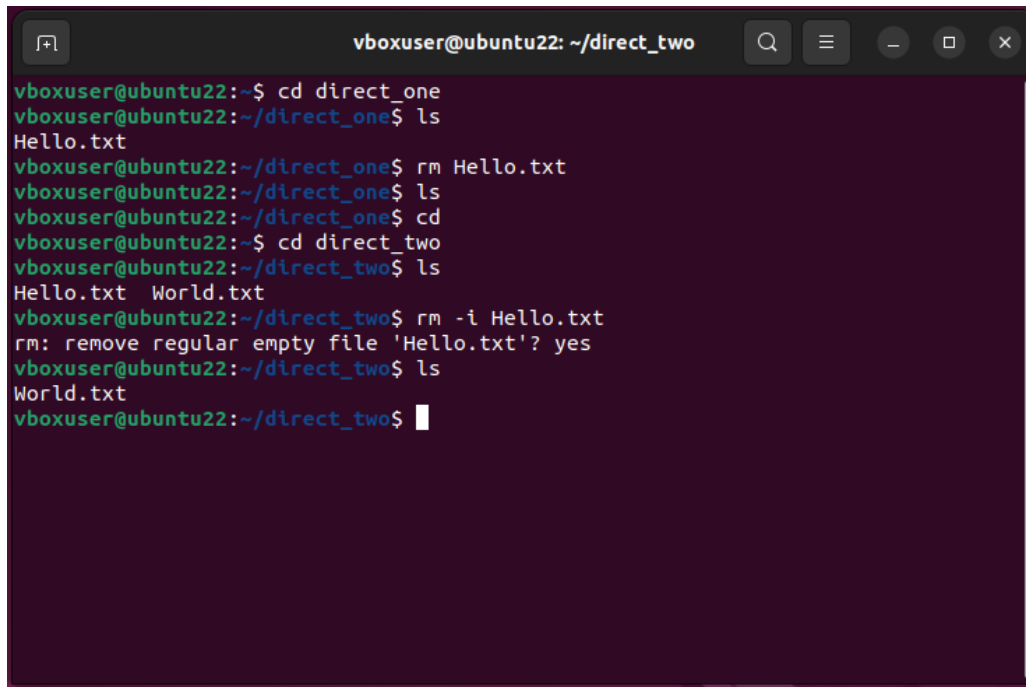
- Remove the first file, as well as the copy you created. <-i important creates prompt prior to deletion> “shred -u file”
 - Removing our first file requires us to ensure we are in direct_one and then using the “rm” command as such: “rm Hello.txt” . Then we can delete the copy by inputting “cd” to return to original directory; followed by “cd direct_two” to enter the second directory so we can delete the copy. Enter “rm Hello.txt”

```

vboxuser@ubuntu22: ~ /direct_one
vboxuser@ubuntu22:~$ cd direct_one
vboxuser@ubuntu22:~/direct_one$ ls
Hello.txt
vboxuser@ubuntu22:~/direct_one$ rm Hello.txt
vboxuser@ubuntu22:~/direct_one$ ls
vboxuser@ubuntu22:~/direct_one$

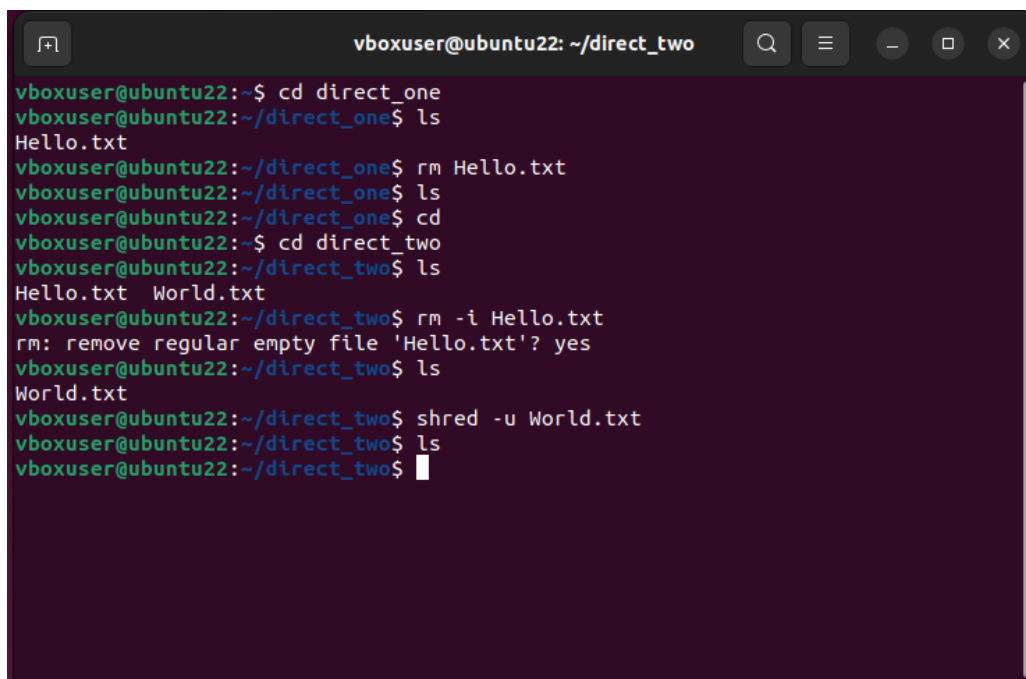
```

- Please be advised that this action cannot be undone. As such, we recommend modifying the command by adding the option “-i” which will create a confirmation prompt prior to deletion.



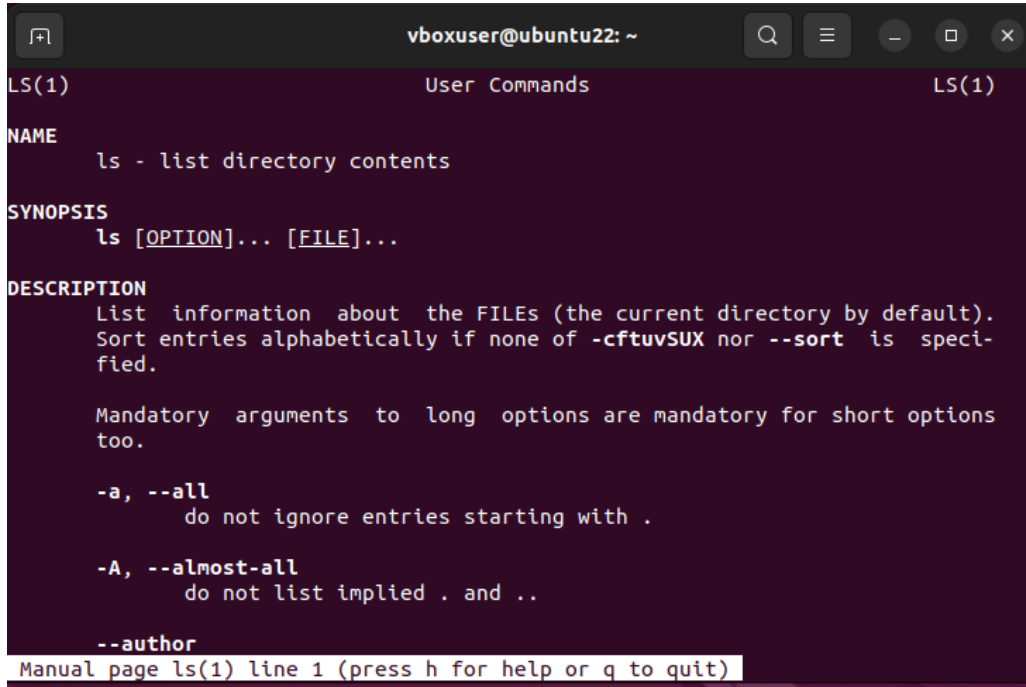
```
vboxuser@ubuntu22: ~/direct_two
vboxuser@ubuntu22:~$ cd direct_one
vboxuser@ubuntu22:~/direct_one$ ls
Hello.txt
vboxuser@ubuntu22:~/direct_one$ rm Hello.txt
vboxuser@ubuntu22:~/direct_one$ ls
vboxuser@ubuntu22:~/direct_one$ cd
vboxuser@ubuntu22:~$ cd direct_two
vboxuser@ubuntu22:~/direct_two$ ls
Hello.txt World.txt
vboxuser@ubuntu22:~/direct_two$ rm -i Hello.txt
rm: remove regular empty file 'Hello.txt'? yes
vboxuser@ubuntu22:~/direct_two$ ls
World.txt
vboxuser@ubuntu22:~/direct_two$
```

- Note: a more secure way to delete sensitive files will require the use of the “shred -u file_name” command which will write over the file multiple times prior to deleting as a safety measure [7].



```
vboxuser@ubuntu22: ~/direct_two
vboxuser@ubuntu22:~$ cd direct_one
vboxuser@ubuntu22:~/direct_one$ ls
Hello.txt
vboxuser@ubuntu22:~/direct_one$ rm Hello.txt
vboxuser@ubuntu22:~/direct_one$ ls
vboxuser@ubuntu22:~/direct_one$ cd
vboxuser@ubuntu22:~$ cd direct_two
vboxuser@ubuntu22:~/direct_two$ ls
Hello.txt World.txt
vboxuser@ubuntu22:~/direct_two$ rm -i Hello.txt
rm: remove regular empty file 'Hello.txt'? yes
vboxuser@ubuntu22:~/direct_two$ ls
World.txt
vboxuser@ubuntu22:~/direct_two$ shred -u World.txt
vboxuser@ubuntu22:~/direct_two$ ls
vboxuser@ubuntu22:~/direct_two$
```

- The manual page for a given command.
 - To display the manual page for a given command, simply enter “man ls” in the terminal from the original directory. When prompted to quit, simply type “q”



A terminal window titled 'vboxuser@ubuntu22: ~' showing the manual page for the 'ls' command. The window has a dark purple background. The text is as follows:

```
LS(1)                                User Commands                                LS(1)

NAME
  ls - list directory contents

SYNOPSIS
  ls [OPTION]... [FILE]...

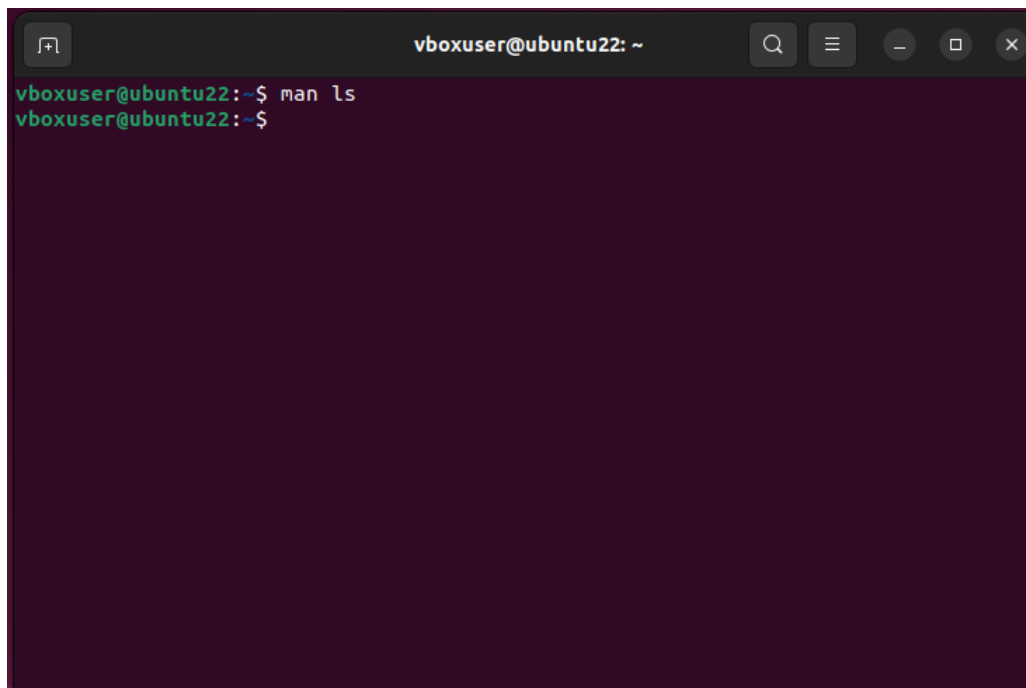
DESCRIPTION
  List information about the FILES (the current directory by default).
  Sort entries alphabetically if none of -cftuvSUX nor --sort is speci-
  fied.

  Mandatory arguments to long options are mandatory for short options
  too.

  -a, --all
        do not ignore entries starting with .

  -A, --almost-all
        do not list implied . and ..

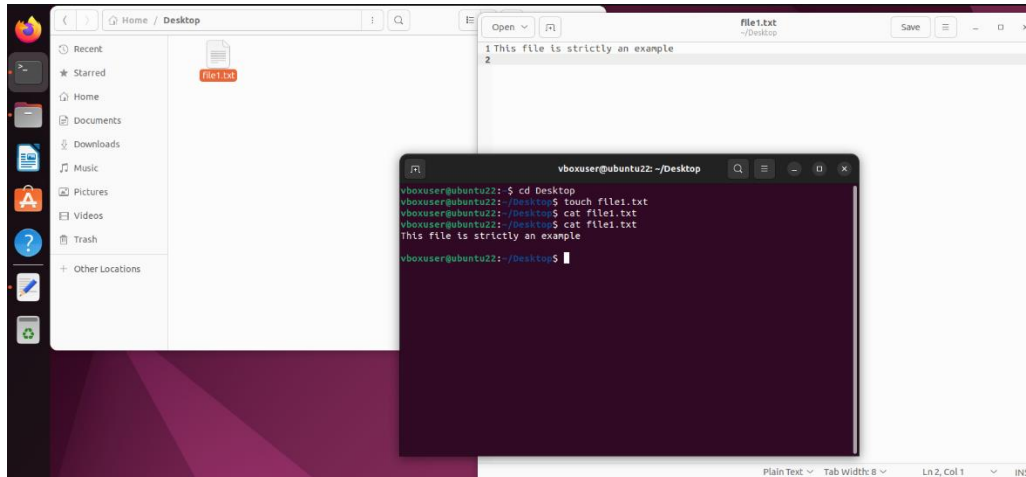
  --author
Manual page ls(1) line 1 (press h for help or q to quit)
```



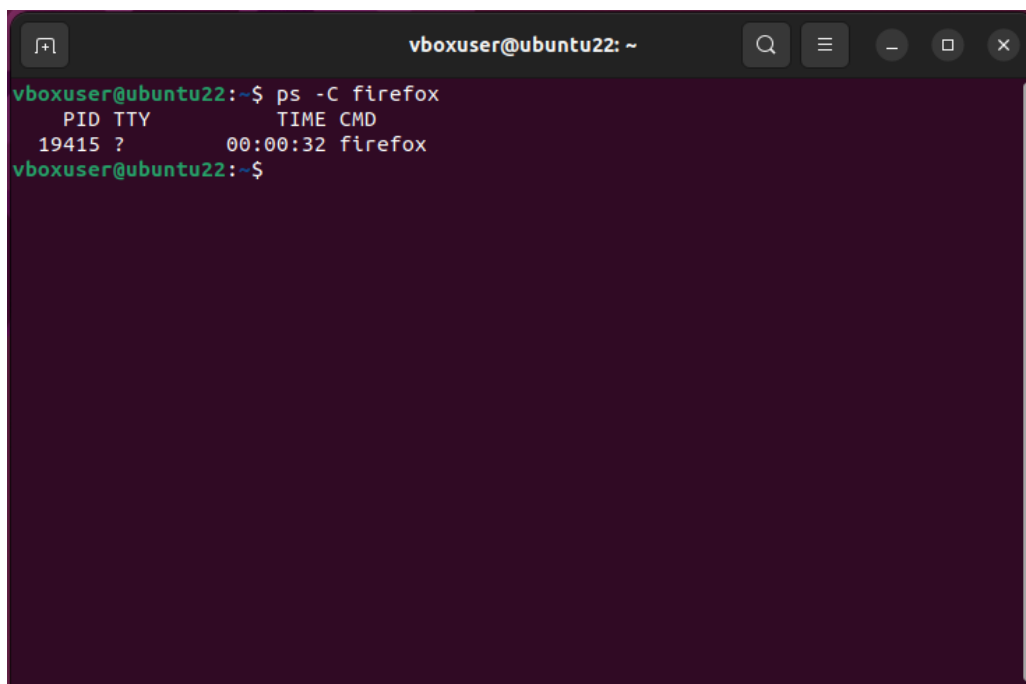
A terminal window titled 'vboxuser@ubuntu22: ~' showing the command 'man ls' being executed. The prompt is 'vboxuser@ubuntu22:~\$'. The output is the manual page for 'ls', which is the same as the one shown in the previous screenshot. The prompt is now 'vboxuser@ubuntu22:~\$' again.

- Create a text file, then use an editor to modify the content. Then display the content of the modified file.

- Here we created a text file through the terminal using the “touch” command and opened that file through the Graphic User Interface (GUI). We made edits through the text editor application. Lastly we used the “cat” command to display everything within that text file on the terminal.

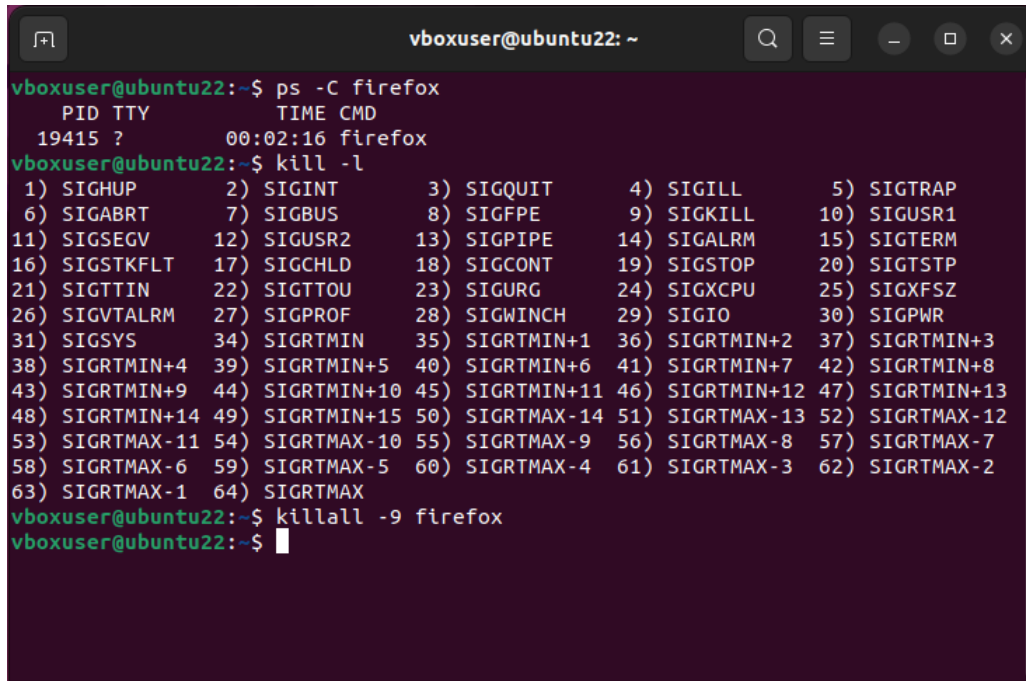


- Show running processes on the system. Demonstrate how to search for a specific process.
 - Here we can show a running process for a specific process/application using the command “ps -C <application>”



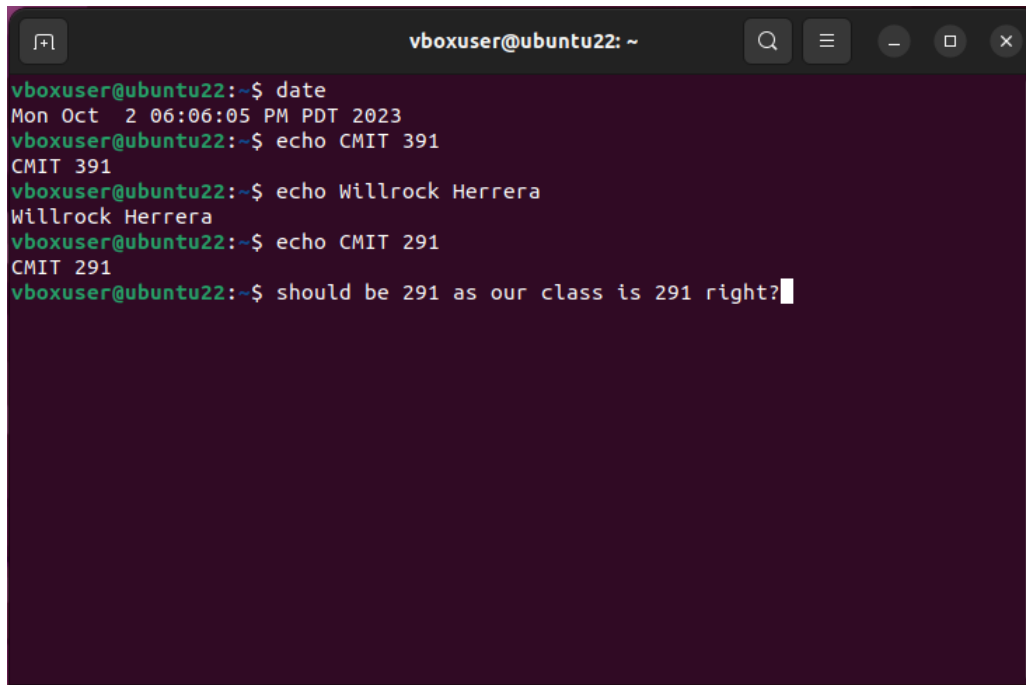
- Forcibly stop a running process with the proper command. Supply screen shot.

- To terminate a process/application from the terminal we will demonstrate using the “kill” command as follows: “kill -l” will display all available signals/options that can be added to the “killall” command [8]. Then we use the “killall” command to terminate all processes with that name. Example: “killall -9 firefox” this example uses the “killall” command with the “-9” using option 9 from “kill -l” to kill signal to process and firefox being the process/application.



```
vboxuser@ubuntu22: ~  
vboxuser@ubuntu22:~$ ps -C firefox  
  PID TTY          TIME CMD  
 19415 ?        00:02:16 firefox  
vboxuser@ubuntu22:~$ kill -l  
 1) SIGHUP      2) SIGINT      3) SIGQUIT     4) SIGILL      5) SIGTRAP  
 6) SIGABRT     7) SIGBUS      8) SIGFPE      9) SIGKILL     10) SIGUSR1  
11) SIGSEGV     12) SIGUSR2     13) SIGPIPE     14) SIGALRM     15) SIGTERM  
16) SIGSTKFLT   17) SIGCHLD    18) SIGCONT     19) SIGSTOP     20) SIGTSTP  
21) SIGTTIN     22) SIGTTOU     23) SIGURG      24) SIGXCPU     25) SIGXFSZ  
26) SIGVTALRM   27) SIGPROF    28) SIGWINCH    29) SIGIO        30) SIGPWR  
31) SIGSYS      34) SIGRTMIN    35) SIGRTMIN+1  36) SIGRTMIN+2  37) SIGRTMIN+3  
38) SIGRTMIN+4  39) SIGRTMIN+5  40) SIGRTMIN+6  41) SIGRTMIN+7  42) SIGRTMIN+8  
43) SIGRTMIN+9  44) SIGRTMIN+10 45) SIGRTMIN+11 46) SIGRTMIN+12 47) SIGRTMIN+13  
48) SIGRTMIN+14 49) SIGRTMIN+15 50) SIGRTMAX-14 51) SIGRTMAX-13 52) SIGRTMAX-12  
53) SIGRTMAX-11 54) SIGRTMAX-10 55) SIGRTMAX-9  56) SIGRTMAX-8  57) SIGRTMAX-7  
58) SIGRTMAX-6  59) SIGRTMAX-5  60) SIGRTMAX-4  61) SIGRTMAX-3  62) SIGRTMAX-2  
63) SIGRTMAX-1  64) SIGRTMAX  
vboxuser@ubuntu22:~$ killall -9 firefox  
vboxuser@ubuntu22:~$
```

Part 3



```
vboxuser@ubuntu22: ~  
vboxuser@ubuntu22:~$ date  
Mon Oct  2 06:06:05 PM PDT 2023  
vboxuser@ubuntu22:~$ echo CMIT 391  
CMIT 391  
vboxuser@ubuntu22:~$ echo Willrock Herrera  
Willrock Herrera  
vboxuser@ubuntu22:~$ echo CMIT 291  
CMIT 291  
vboxuser@ubuntu22:~$ should be 291 as our class is 291 right?
```

Conclusion

To reiterate the value of this Operating System, this migration is full of benefits for this organization and will allow Faster Computing to prosper as it should. It will be easy to use for all with its user-friendly GUI and several compatible and alternative applications. For my higher ups, you will be able to monitor and control access and permissions for all users on the network easily and your IT department will love the ease-of-use terminal for full control. The company will now have extra funds to reallocate for anything else it may need like improvements or expansion ideas. Again, thank you so much for allowing us to join you on this amazing adventure! Cheers to this move and cheers to our future endeavors!

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